Qian Qian

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Skills

- **Languages**: Java, Kotlin, Python, Rust, C, C++, Shell, Swift, Go, Typst, LaTeX
- **Tech/Tools**: Git, GitHub Actions, Linux, Unit Testing, JUnit, PyTest, NGINX, Docker, Ktor, Flask, SwiftUI, Quilt

Experience

Software Developer & Software Subsystem Co-Lead Sep 2025 — Present
Waterloo Rocketry *Waterloo, Ontario, Canada*

- Developed Mission Control software supporting the **world's highest-altitude amateur liquid rocket launch**.
- Developed Python-based data acquisition and processing systems with PyTest coverage for reliable Mission Control telemetry and built Docker/Ansible deployment infrastructure for consistent, reproducible environments.
- Assembled and tested ground-side electrical and sensor systems and served as a Mission Control data acquisition (DAQ) operator, supporting launch-pad operations during integration and pre-launch testing.
- Lead a team of ~20 developers in designing and developing Mission Control and ground support systems, mentoring developers in engineering practices including design reviews, unit testing, code reviews, and collaborative debugging.

Software Engineering Intern, Build and Automation May 2026 — Aug 2026
Maxon *Montreal, Quebec, Canada*

- Built a Redshift rendering performance-monitoring system to identify regressions across standard rendering unit tests using Python, Grafana, and Loki for data processing, aggregation, and visualization.
- Automated ArchVis changelog generation and build notifications using Python and the GitLab and Jira APIs, giving QA visibility into resolved issues to guide targeted testing.

Projects

Bitarium - *Rust, Zig, OCaml, RISC-V* Oct 2026 — Present

- Built a modular computer emulator with separate CPU, memory, and device subsystems, using Rust for execution, Zig for memory and devices, and OCaml for assembly tools.
- Integrated Rust and Zig components and validated CPU behavior against an independent OCaml reference model.

QubeC - *Rust, Zig, Kotlin, C, C++, Go* Aug 2025 — Present

- Built a bytecode compiler and stack-based virtual machine for a custom C-like programming language.
- Developed experimental projects in multiple languages to explore language features and compare syntax, tooling, and programming paradigms.

Typstify - *Swift, SwiftUI, Rust* Jul 2024 — Present

- Developed a native Typst editor for iPadOS to support document authoring and preview workflows using SwiftUI with a Rust-based Typst backend, bridged via swift-bridge for cross-language interoperability.
- Contributed patches to 3 upstream dependencies to expand functionality and resolve existing issues.

Maintainer, ungoogled-chromium-macos - *Shell, Python, Build Systems* Nov 2023 — Present

- Maintained the macOS distribution of ungoogled-chromium, a privacy-focused browser, delivering 100+ releases to 100,000+ users through upstream tracking, build-script maintenance, and downstream patch management.
- Reduced CI build time by 40% by optimizing GitHub Actions workflows to accelerate build time and release cycles.

CalcLox - *Java* Jan 2025 — May 2025

- Designed and implemented a Lox-based scripting language and its interpreter for a programmable calculator.
- Built JUnit unit tests to validate language semantics and interpreter correctness across core calculator workflows.

Education

University of Waterloo Waterloo, Ontario, Canada
Candidate for Bachelor of Software Engineering, Honours Software Engineering, Co-op *Sep 2025 — May 2030*